

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Statement 1 | Every element of a group generates a cyclic subgroup of order 10 if the group has 10 elements.' is 'True, True'.
2. True or False: The correct answer to 'Statement 1 | If H is a subgroup of a group G and a belongs to H , then a^{-1} also belongs to H .' is 'True, True'.
3. True or False: The correct answer to 'Find the product of the given polynomials in the given polynomial ring. $(x^2 + 1)(x^2 + 2)$ in $\mathbb{Z}_3[x]$ ' is ' $x^4 + 2x^2 + 1$ '.
4. True or False: The correct answer to 'Let V and W be 4-dimensional subspaces of a 7-dimensional vector space V . Then $V \cap W$ is 3-dimensional.' is 'True, True'.
5. True or False: The correct answer to 'Statement 1 | If a and b are elements of finite order in an Abelian group, then $\text{lcm}(\text{order}(a), \text{order}(b))$ divides $\text{order}(ab)$.' is 'True, True'.
6. True or False: The correct answer to 'Compute the product in the given ring. $(20)(-8)$ in $\mathbb{Z}_{26}$ ' is '22'.
7. True or False: The correct answer to 'Statement 1 | Every ideal in a ring is a subring of the ring.' is 'True, True'.
8. True or False: The correct answer to 'Statement 1 | A factor group of a non-Abelian group is non-Abelian.' is 'True, True'.
9. True or False: The correct answer to 'Statement 1 | $\mathbb{R}$ is a splitting field of some polynomial over $\mathbb{Q}$.' is 'True, True'.
10. True or False: The correct answer to 'How many positive numbers x satisfy the equation $\cos(97x) = \cos(x)$?' is '1'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Find the integer closest in value to

$$\frac{\sqrt[4]{\tan 60^\circ} \sqrt[4]{\tan 30^\circ}}{\sqrt[4]{\tan 45^\circ}}$$

2. A function f is defined by $f(z) = \frac{1}{z^2}$ for $z \neq 0$. Find the residue of f at $z = 0$.

End of Paper